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M72/AS01E tuberculosis vaccine: how dose and regimen were selected

Bottom line

The documentary record supports Tom Evans's distinction between **dose selection** and **schedule selection**. The final **10 µg M72/AS01E dose was the product of a genuine human dose/formulation study**: 10 and 20 µg M72 with the lower-adjuvant AS01E formulation were compared with the previously used 40 µg M72/AS01B formulation, and 10 µg M72/AS01E was selected because it produced similar M72-specific CD4 T-cell responses with less antigen and less adjuvant. The trial was not powered as a formal equivalence study, but this was a prospectively specified dose-selection exercise. [1]

By contrast, I found **no published human M72 trial that randomised the interval between dose 1 and dose 2**. The one-month interval was already used in early M72 formulation studies and was then retained in dose-finding, studies across different populations, Phase IIb, and now Phase III. The evidence therefore establishes that 0–28/30 days is immunogenic and efficacious, but does **not** establish that it is close to the optimal interval. The present Phase III trial again gives the two doses on approximately days 1 and 29 rather than testing an alternative interval. [2]

A **0–56-day regimen is scientifically plausible but currently unproven for M72**. There is some supportive evidence from AS01-adjuvanted vaccines and limited preclinical M72 evidence for delayed boosting, but nothing that substantiates Evans's more quantitative claim that efficacy might improve by around 20%. The ongoing M72 correlates-of-protection programme could make schedule optimisation substantially easier, but only if it identifies and validates an immune marker that predicts protection rather than merely vaccine exposure. WHO states that Phase IIb samples are being analysed for candidate correlates and that Phase III is intended to evaluate and potentially validate them. [3]

How the final dose was reached

M72 is the stabilised successor to the Mtb72F fusion antigen. In an early head-to-head human study, 110 Mtb-naïve adults received **40 µg M72/AS01, 40 µg M72/AS02, 40 µg Mtb72F/AS02, 40 µg unadjuvanted M72, or AS01 alone**, on a **0- and 1-month schedule**. M72/AS01 generated stronger CD4 responses than M72/AS02, while changing Mtb72F to M72 did not itself explain the improvement with AS02. This provided direct evidence for choosing the AS01 platform, but not for the one-month interval. [4]

The decisive dose-finding trial was NCT00621322 in PPD-positive Filipino adults. Participants received two injections one month apart of **40 µg M72/AS01B, 20 µg M72/AS01E, 10 µg M72/AS01E, 10 µg M72/AS02D, 40 µg M72/saline, or AS01B alone**. AS01E contained half the immunostimulant quantity of AS01B. The prespecified objective was unusually explicit for vaccine development: select the **lowest M72/AS01 dose** that remained clinically acceptable and gave CD4 responses comparable with the previously studied 40 µg M72/AS01B formulation. [1]

At one month after the second vaccination, the three AS01 groups were not significantly different on the principal M72-specific CD4 endpoint: 10 µg AS01E versus 20 µg AS01E, p=0.18; 10 µg AS01E versus

40 µg AS01B, *p*=0.31; and 20 µg AS01E versus 40 µg AS01B, *p*=0.68. The 10 µg AS01E formulation was superior to 10 µg M72/AS02D on that endpoint. The investigators therefore selected **10 µg M72/AS01E**. Importantly, they explicitly cautioned that the study was not designed as a formal equivalence trial; the evidence is best described as a fairly good dose/formulation-selection experiment rather than proof that 10 µg is biologically optimal. [1]

Subsequent studies established that two doses generated robust responses in adolescents, adults with M. tuberculosis infection, people living with HIV and BCG-vaccinated infants. In the infant study, two M72/AS01 doses generated higher humoral and CD4 responses than one dose, giving some direct support for **two doses rather than one**. None of these studies, however, appears to have varied the interval experimentally. The Phase IIb efficacy trial consequently took forward the same two-dose, one-month regimen. It produced 54.0% efficacy at the primary analysis and 49.7% efficacy over three years. [5]

Development question	Direct M72 evidence	What was learned
M72 versus predecessor/ formulation	40 µg formulations, AS01 versus AS02 and controls	AS01 gave stronger CD4 responses. [4]
Antigen/adjuvant dose	10 µg AS01E, 20 µg AS01E, 40 µg AS01B plus controls	Little evidence of additional CD4 response across these AS01 formulations; 10 µg AS01E selected as lowest acceptable formulation. [1]
One versus two doses	Direct evidence in infants and boosting in adult studies	Second dose generally increased immune responses. [6]
One-month versus longer interval	**No human interval-randomised M72 study identified**	One month is proven workable, not proven optimal.
Clinical efficacy	Two doses one month apart	Approximately 50% protection over three years in IGRA-positive adults. [7]

The resulting asymmetry is quite sharp: **antigen/adjuvant dose was actively de-escalated; the interval was essentially held fixed while the rest of the product was optimised.**

What supports, and does not support, the 0–28-day interval

There is a reasonable empirical justification for *using* the one-month schedule. In the early formulation comparison it generated strong responses; in the dose-finding study the second vaccination boosted responses for the selected AS01E formulations; and eventually the same regimen demonstrated clinical efficacy in Phase IIb. [8] What is absent is evidence answering the optimisation question: **would the same two injections work better at 0–56, 0–84, or another interval?**

The development sequence strongly suggests path dependence. The 0,1-month cadence was already fixed in the 40 µg formulation-comparison study; the formal dose-finding trial altered antigen and adjuvant quantity while holding that cadence constant; later studies inherited it; and Phase IIb then supplied the first efficacy evidence for precisely that regimen. [8] Once the Phase IIb regimen had demonstrated approximately 50% efficacy, retaining it for confirmatory Phase III had an obvious

evidential advantage, even though that does not answer whether another interval would have been superior.

There is one intriguing piece of developer documentation: a later GSK patent describes M72 regimens with substantially longer intervals and includes a mouse experiment in which a **delayed later M72/AS01 immunisation produced stronger antibody and polyfunctional CD4 responses than the shorter schedule**. The patent also claims M72-related regimens with first-to-second-dose intervals extending from roughly two months upwards. [9] This establishes that delayed boosting was biologically contemplated within the M72 programme. It is weak evidence for 0–56 in humans: the supporting experiment involved mice and a different multi-dose design, and patent claims are not estimates of clinical efficacy.

Evidence bearing on longer spacing

Related AS01 vaccines make Evans's hypothesis plausible, but do not establish it. For recombinant zoster vaccine, a randomised trial in adults aged at least 50 compared two gE/AS01B doses **2, 6 or 12 months apart**. The six-month regimen met the study's antibody non-inferiority criterion relative to the standard two-month regimen, while the 12-month regimen did not meet that particular criterion despite high vaccine-response rates. Thus AS01-mediated boosting plainly remains effective after much longer intervals; the study does not show a monotonic advantage from increasing the interval. [10]

RTS,S/AS01 malaria work gives a stronger proof of principle that a delayed later immunisation can materially change the quality and magnitude of immunity. Delayed/fractional third-dose regimens have produced improved immune profiles and, in controlled malaria-infection experiments, encouraging protection. However, several of these experiments changed **dose and timing together**, making the pure interval effect difficult to identify. Extrapolation is further limited because RTS,S protection relies importantly on antibodies against malaria sporozoites, whereas the protective mechanism for M72 remains unknown. [11]

For M72 specifically, Phase IIb showed that M72-specific antibody concentrations and CD4 frequencies rose after vaccination and persisted for years, but neither is presently established as the mediator or validated correlate of protection. [12] General arguments about allowing additional time for maturation of immune memory therefore supply **mechanistic plausibility**, not a quantitative prediction for TB efficacy.

Accordingly, the evidence grades roughly as follows: **direct evidence that 0–56 improves M72 efficacy: none identified; direct M72 immunogenicity evidence comparing 0–28 with 0–56: none identified; M72 preclinical evidence favouring delayed boosting: limited; related-AS01 evidence that longer intervals can work well and sometimes alter immunity favourably: moderate; basis for a claim such as “20% better efficacy”: not identified.**

Correlates of protection and future optimisation

The Phase IIb programme was unusually valuable for correlates work because almost all participants consented to biobanking before and after vaccination. The original efficacy publication explicitly identified those samples as an opportunity to discover immune correlates of vaccine-mediated protection. [13] WHO now says that a **comprehensive analysis of Phase IIb samples is under way**, and that Phase III should permit further evaluation and potential validation of candidate correlates. WHO continues to state that the mechanism of M72-mediated protection is unknown, despite established induction of IFN-γ-producing CD4 T cells and antibodies. [3]

This aligns with Richard White's account in your interview notes that the Phase IIb correlates effort has been running for several years. Publicly described work includes multi-platform analysis of stored material intended to integrate proteomic, metabolomic, lipidomic and transcriptomic information. At present, I found **no publicly validated M72 correlate that can be used as a surrogate efficacy endpoint**. Recent systems-biology analyses identifying vaccine-associated signatures should not be read as having solved this problem: separating vaccine recipients from controls immunologically is much easier than distinguishing protected from breakthrough cases conditional on vaccination.

A validated correlate could change the optimisation problem considerably. Instead of requiring separate TB-disease efficacy trials for 0–28 and 0–56 schedules, a comparatively small randomised immunogenicity study could compare dose and interval combinations against a validated protective marker. But there are two important thresholds. First, the marker must predict **individual protection within vaccinated people**, rather than merely respond to vaccination. Second, evidence should establish that changing the marker through dose or schedule is informative about changing protection; otherwise optimisation against it risks selecting a regimen that moves the biomarker without improving efficacy. Phase III provides the large independent dataset needed to test the first of these propositions. WHO explicitly envisages such validation. [3]

Published record versus the interview claims

Your interview notes are broadly consistent with the documentary record on the main qualitative point. Evans's statement that **M72 dose development was "not too bad"** is supported: there was a purposeful human comparison spanning a four-fold antigen range and two AS01 adjuvant levels, and it resulted in downward selection to 10 µg M72/AS01E rather than simply carrying forward the largest tolerated formulation. [1]

His criticism of the **0–28 schedule** also identifies a real evidential gap. I found no documentary evidence that 0–28 was selected because it defeated 0–56 or another interval in an M72 comparison. The published record instead shows repeated inheritance of the same interval from early formulation studies through efficacy trials. [2] Calling that interval a "mistake", however, goes beyond the evidence: it is a regimen with demonstrated efficacy, while its counterfactual alternatives have not been tested.

Finally, Evans's suggestion that **0–56 might have produced substantially greater efficacy, including the approximately 20% figure recorded in your notes, remains expert speculation**. The direction is biologically credible and related AS01 evidence makes a longer-interval trial quite defensible, but I found no M72 clinical comparison, validated correlate, or M72-specific model that quantifies such an efficacy gain. The strongest defensible conclusion is therefore narrower: **M72's antigen/adjuvant dose received meaningful optimisation; its one-month interval apparently did not, and 0–56 remains a plausible but empirically unresolved alternative.**

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[1] A randomized, controlled dose-finding Phase II study of the M72/AS01 candidate tuberculosis vaccine in healthy PPD-positive adults - PubMed
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[2] [4] [8] Improved CD4⁺ T cell responses to Mycobacterium tuberculosis in PPD-negative adults by M72/AS01 as compared to the M72/AS02 and Mtb72F/AS02 tuberculosis candidate vaccine formulations: a randomized trial.
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